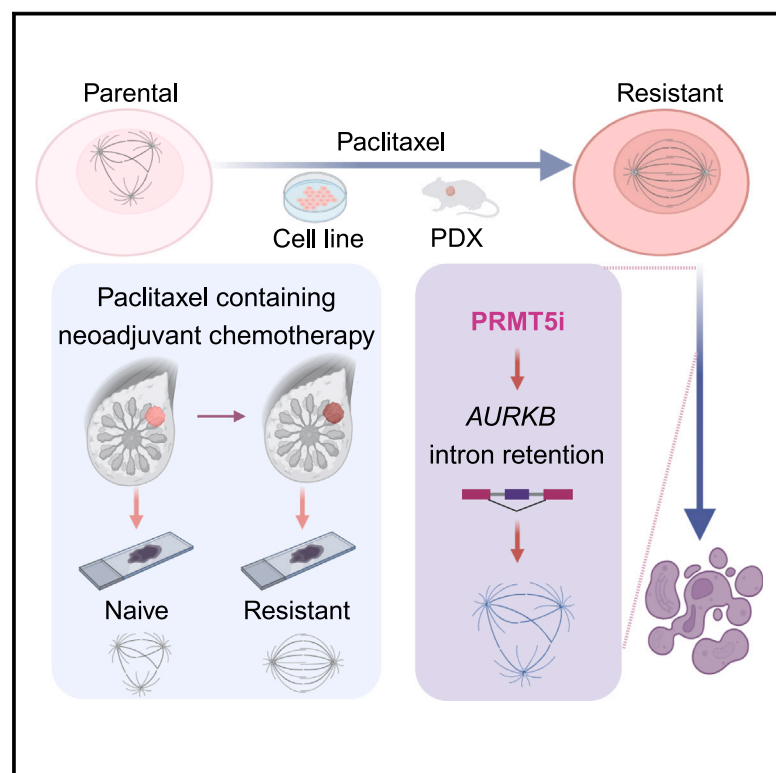


Cell Chemical Biology

A chemical screen identifies PRMT5 as a therapeutic vulnerability for paclitaxel-resistant triple-negative breast cancer

Graphical abstract



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In brief

Paclitaxel-resistant triple-negative breast cancer (TNBC) remains challenging to treat. Zhang et al. discover that resistant TNBC cells are vulnerable to PRMT inhibition, which decreases AURKB protein levels and induces mitotic catastrophe. Inhibiting type I PRMTs and PRMT5 synergizes to suppress tumor growth, suggesting a promising strategy for treating paclitaxel-resistant TNBC.

Highlights

- Paclitaxel-resistant triple-negative breast cancer is vulnerable to PRMTs inhibition
- Inhibiting type I PRMTs and PRMT5 synergizes to suppress paclitaxel-resistant TNBC growth
- Targeting PRMTs induces intron retention in AURKB, leading to mitotic catastrophe



Article

A chemical screen identifies PRMT5 as a therapeutic vulnerability for paclitaxel-resistant triple-negative breast cancer

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SUMMARY

Paclitaxel-resistant triple negative breast cancer (TNBC) remains one of the most challenging breast cancers to treat. Here, using an epigenetic chemical probe screen, we uncover an acquired vulnerability of paclitaxel-resistant TNBC cells to protein arginine methyltransferases (PRMTs) inhibition. Analysis of cell lines and in-house clinical samples demonstrates that resistant cells evade paclitaxel killing through stabilizing mitotic chromatin assembly. Genetic or pharmacologic inhibition of PRMT5 alters RNA splicing, particularly intron retention of aurora kinases B (AURKB), leading to a decrease in protein expression, and finally results in selective mitosis catastrophe in paclitaxel-resistant cells. In addition, type I PRMT inhibition synergies with PRMT5 inhibition in suppressing tumor growth of drug-resistant cells through augmenting perturbation of AURKB-mediated mitotic signaling pathway. These findings are fully recapitulated in a patient-derived xenograft (PDX) model generated from a paclitaxel-resistant TNBC patient, providing the rationale for targeting PRMTs in paclitaxel-resistant TNBC.

INTRODUCTION

Triple-negative breast cancer (TNBC) is characterized by the absence of estrogen receptors (ERs) and progesterone receptors (PRs), as well as the absence of human epidermal growth factor receptor 2 (HER2) gene amplification.^{1,2} Although approximately 30% of TNBC cases are characterized by low levels of HER2 expression and have shown responsiveness to anti-HER2 therapies, chemotherapy remains a mainstay for the standard therapy.^{3,4} Paclitaxel is the principal chemotherapeutic regimen for early-stage or metastatic TNBC patients.⁵ The combination of paclitaxel with immune checkpoint blockade has significantly prolonged the survival of advanced TNBC patients.^{6,7} Despite these advances, resistance arises in around 5–7 months after treating with a paclitaxel-containing regimen

(IMpassion 130 clinical trial),⁷ leading to rapid tumor progression and poor prognosis. Therefore, identifying new therapeutic targets to eradicate paclitaxel-resistant TNBCs is a critical unmet clinical need.

Paclitaxel stabilizes microtubules in cells, preventing their depolymerization and disrupting the normal dynamic equilibrium of microtubule assembly and disassembly during cell division. This disruption ultimately leads to cell-cycle arrest and apoptosis in rapidly dividing cells, making paclitaxel an effective chemotherapeutic agent for various cancers, including breast cancer.⁸ However, recent research has proposed a new perspective, indicating that the clinical dosage of paclitaxel may be insufficient to solely prevent microtubule depolymerization. Instead, it may induce the formation of multipolar spindles in cancer cells, leading to abnormal cell division and subsequent cell death.⁹



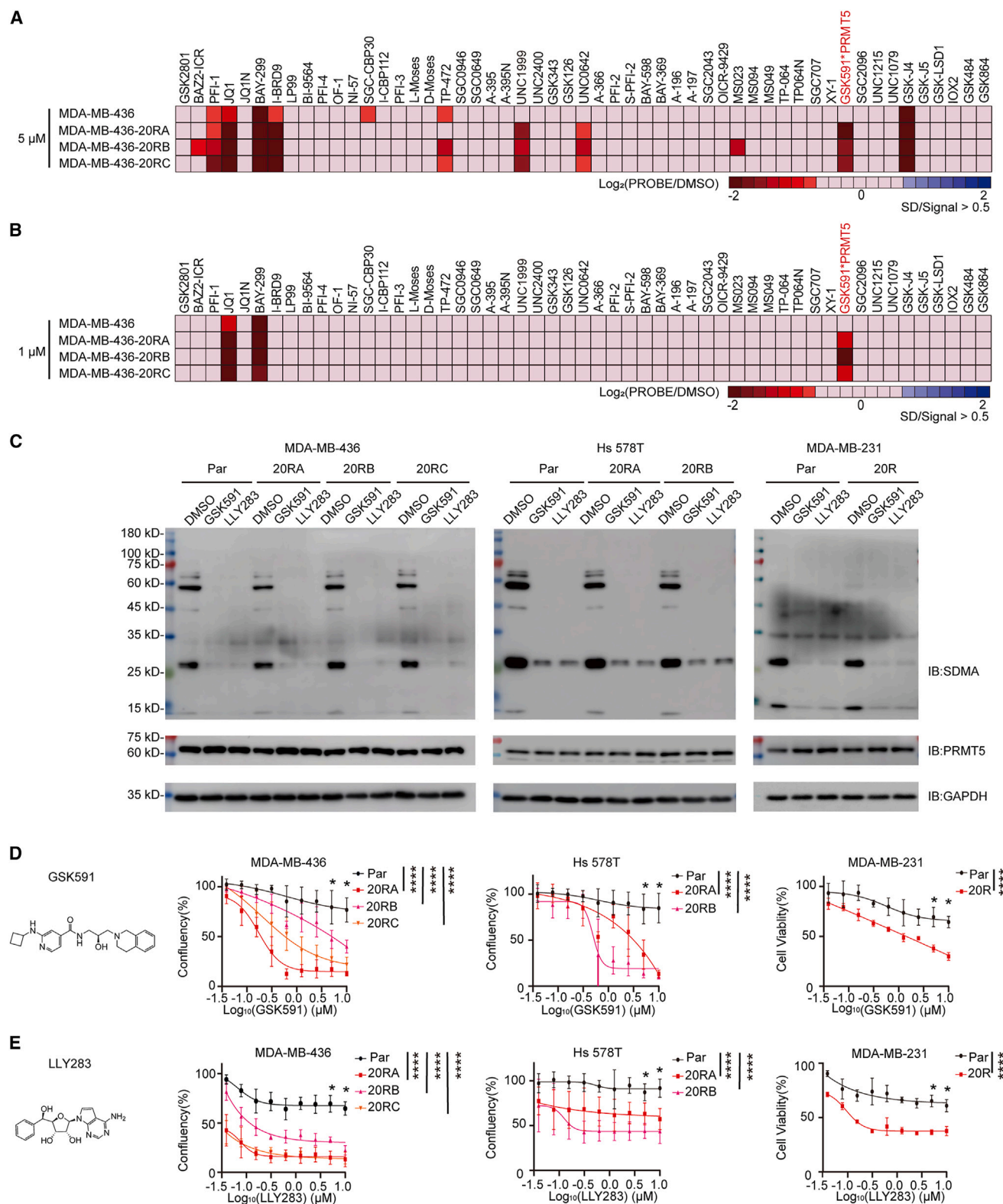


Figure 1. PRMT5 is a therapeutic target for paclitaxel-resistant TNBC

(A and B) Heatmap showing average cell confluence of parental (MDA-MB-436) and paclitaxel-resistant cell lines (MDA-MB-436-20R(A/B/C)) treated with epigenetic chemical probes at (A) 5 μ M or (B) 1 μ M for 5 days treatment (data are shown as mean $\pm$ SD of $n = 4$).

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Additionally, repression of transposable elements's expression by chromatin variants has been associated with paclitaxel-resistant TNBCs, promoting tumor growth by epigenetically repressing the viral mimicry response.¹⁰ These findings reveal the complex mechanisms of paclitaxel resistance and suggest a potential vulnerability to epigenetic regulators. However, the systematic identification of epigenetic therapeutic targets to overcome this resistance remains elusive.

Here, with epigenetic tool compound library screening, we discover that paclitaxel-resistant cells acquire vulnerability to PRMT5 (protein arginine methyltransferase 5) inhibitor. Mechanistically, PRMT5 inhibition perturbs RNA splicing to impair chromatin stability, which is required for survival of resistant cells and thus induces mitotic catastrophe. In addition, inhibition of type I PRMTs in combination with PRMT5 inhibition exerts a synergy that result in repressing tumor growth by further promoting chromatin instability. These findings are validated in immortalized cell lines, PDX (patient-derived xenograft), and clinical samples. Collectively, our data indicate that targeting PRMTs that regulate RNA splicing and chromatin instability is a promising approach for overcoming paclitaxel resistance and improving the survival of TNBC patients.

RESULTS

Paclitaxel-resistant TNBCs acquire vulnerability to PRMT5 inhibition

To identify new therapeutic targets for paclitaxel-resistant TNBC, we first constructed six independent drug resistant derivatives of three TNBC cell lines, MDA-MB-436-20R(A/B/C), Hs 578T-20R (A/B) (A, B, and C indicate the different resistant cell clones), and MDA-MB-231-20R from their parental lines MDA-MB-436, Hs 578T, and MDA-MB-231, respectively, by long-term culture with incrementally escalating doses of paclitaxel¹⁰ (Figure S1A). Cell viability assays confirmed the reduced sensitivity of these resistant cell lines to paclitaxel treatment (Figure S1B). Notably, while the resistant strain of Hs 578T cell line demonstrated a tendency toward growth arrest and cell-cycle arrest in the S phase, such effects were inconsistently observed in the paclitaxel-resistant MDA-MB-436 and MDA-MB-231 cell lines (Figures S1C and S1D). These results suggest that while diverse rates of cell growth and cycle progression may occur during the establishment of paclitaxel-resistant strains, they may not necessarily constitute the primary drivers of resistance in these cells.

The epigenetic compound library comprising 53 probes was then exploited to systematically identify the cellular vulnerabilities associated with paclitaxel resistance (Table S1). This particular library had been previously employed in screenings across various cancer models, effectively revealing new therapeutic targets for cancer treatment.^{11–13} Probes resulting in more than a 2-fold reduction in cell growth compared to DMSO controls in

resistant cell lines, while showing no such effect in parental lines, and were considered as potential agents exhibiting acquired vulnerability in paclitaxel-resistant cell lines.

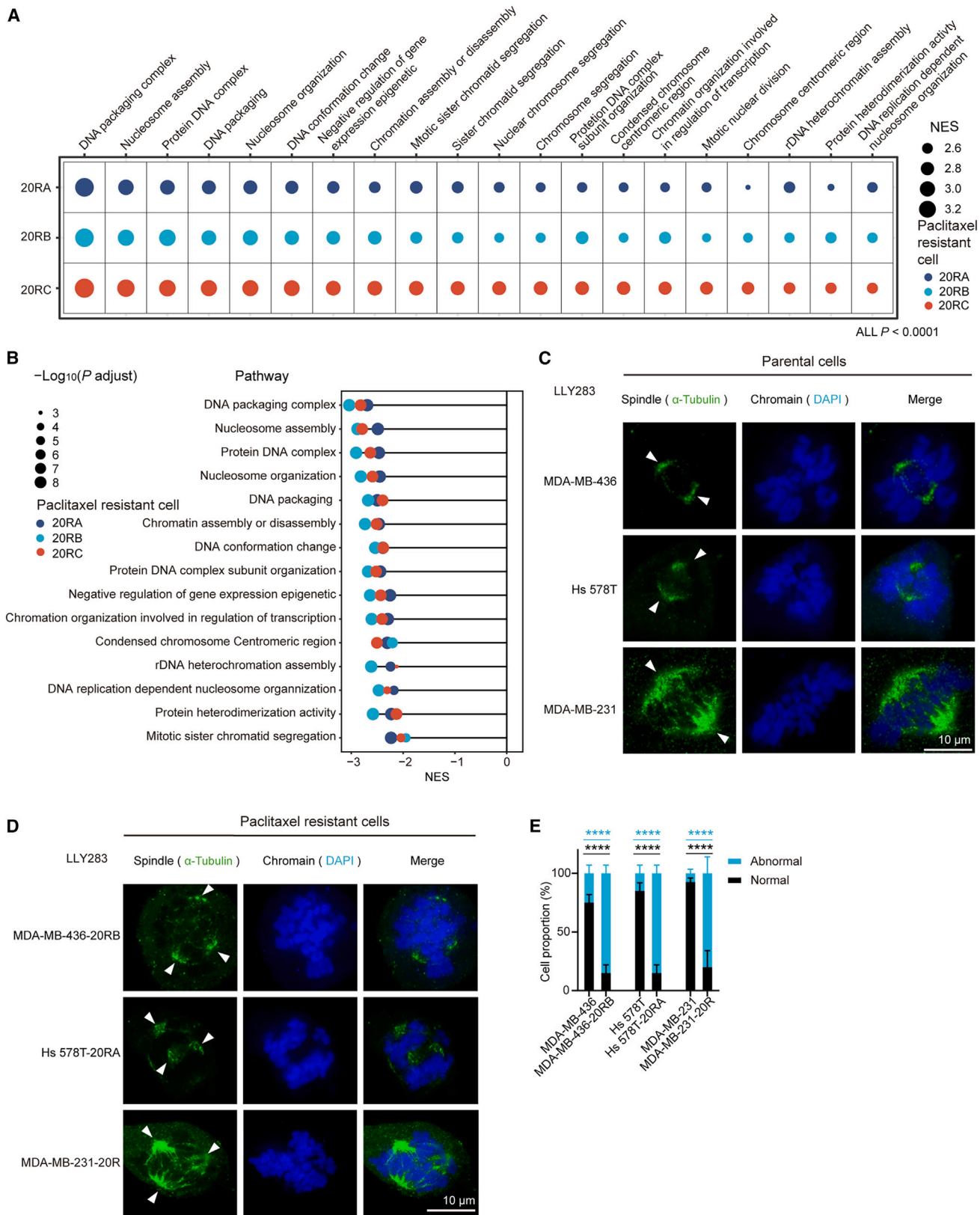
To ensure comprehensive screening of inhibitors across different drug-resistant clones, we opted to include both the parental MDA-MB-436 cells and their corresponding drug resistant clones, in our initial probe screening. The anti-proliferation effects of these probes were determined based on the cell confluency after 5 days treatment using a live-cell imaging platform (Table S2). Among these probes, BAY-299, a dual-activity chemical probe targeting both BRPF2 and TAF1, displayed remarkable anti-tumor effects in both parental and paclitaxel-resistant cells.¹⁴ These data suggest that BRPF2 and/or TAF1 may exert a substantial inhibitory effect on TNBC, regardless of their paclitaxel-resistant or parental nature. Additionally, five compounds, including JQ1, I-BRD9, UNC1999, UNC0642, and GSK591, exhibited significantly increased sensitivity in all three resistant cell lines at a concentration of 5 μ M (Figures 1A and S1E). Among these, the EZH2 inhibitor UNC1999 and BET and BRD family inhibitors JQ1 and I-BRD9 had been previously reported for their effectiveness in suppressing tumor growth in paclitaxel-resistant TNBC.^{10,15,16} Compared to the G9a inhibitor UNC0642, PRMT4 inhibitor MS049, PRMT3 inhibitor SGC707, and type I PRMT inhibitor MS023, PRMT5 inhibitor GSK591 displayed enhanced sensitivity in resistant cells by significantly impairing cell growth at a lower concentration of 1 μ M (Figures 1B and S1F). Therefore, GSK591 was selected as a candidate agent for further study.

PRMT5 is a type II arginine methyltransferase that catalyzes symmetric dimethylation of arginine (SDMA) on proteins involved in transcription, RNA splicing, DNA damage repair, and immune response regulation. Remarkably, PRMT5 is currently being investigated in eleven clinical trials for its potential in cancer therapy.¹⁷ To confirm the phenotype observed in the screening assay, we evaluated the anti-proliferative effects of two PRMT5 inhibitors with distinct mechanisms of action, GSK591 (a substrate competitive inhibitor)¹⁸ and LLY283 (a SAM competitive inhibitor),¹⁹ on three parental cells and their six paclitaxel-resistant derivatives (Table S1). These PRMT5 inhibitors reduced the levels of SDMA in both parental and resistant lines but have no effect on PRMT5 expression (Figures 1C, S1G, and S1H). Paclitaxel-resistant cell lines exhibited dose-dependent sensitivity to both PRMT5 inhibitors (Figures 1D and 1E). The on-target effect of PRMT5 was further validated through genetic knock-down and rescue assays in two resistant cell lines but not for parental cells (Figures S1I–S1M).

Although the response to GSK591 varied among different resistant cell lines derived from the same parental cell, all exhibit heightened sensitivity compared to their parental counterparts, highlighting PRMT5 as a promising target for overcoming paclitaxel resistance (Figures 1D and 1E; Table S2). This variability in response is likely attributed to inherent heterogeneity among drug-resistant clones, potentially stemming from differences in

(C) Immunoblots of SDMA and PRMT5 in parental and paclitaxel-resistant cells after 5 days treatment with 1 μ M GSK591, LLY283, or DMSO control. GAPDH was used as a loading control. Images are representatives of three independent biological replicates.

(D and E) Cell viability determined by the cell counting kit-8 (CCK-8) assay or confluence curves of parental cells (MDA-MB-436, Hs 578T, and MDA-MB-231) and paclitaxel-resistant cells (MDA-MB-436-20R(A/B/C), Hs 578T-20R(A/B), and MDA-MB-231-20R) treated with (D) GSK591 or (E) LLY283 for 5 days. The concentration of the PRMT5 inhibitors ranged from 10 μ M and subsequently diluted 2-fold to 0.039 μ M, resulting in a total of 9 different concentrations (mean $\pm$ SD, $n = 3$, Student's t test, * indicates the concentrations used for p values calculation). **** $p < 0.0001$.



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RNA expression, epigenetic states, or protein expression. At high doses of the PRMT5 inhibitor, the growth inhibition curve reached a plateau, which is likely due to the presence of a resistant cell population to PRMT5 inhibition, thus warranting exploration of another synergistic target. (Figures 1D and 1E).

PRMT5 inhibition induced chromosomal instability in paclitaxel-resistant cells

To investigate the molecular basis of the selective dependency on PRMT5, we first compared the expression level of PRMT5 in parental and resistant cell lines. Neither mRNA expression (Figure S2A) nor protein level showed a significant difference (Figures 1C and S2B). No significant alterations of SDMA levels were detected (Figure S1G). To further understand how PRMT5 activity impacts paclitaxel resistance, transcriptome analysis was performed on MDA-MB-436 and MDA-MB436-20R (A/B/C) cells. Compared to the parental lines, gene set enrichment analysis (GSEA) revealed that most of the top 20 pathways enriched in three resistant clones were associated with chromosomal assembly and segregation during mitosis (Figure 2A; Table S3). Key genes associated with chromosomal assembly and segregation signaling pathways, such as *H1-3*, *H2BC5*, *H2BC21*, and *H3C2* were further validated through real time qPCR experiments (Figures S2C and S2D).^{20,21} No consistent upregulation in pathways associated with drug pumps or stem cell markers in resistant cells was observed (Figures S2E–S2G). Given a previous report demonstrating that paclitaxel induces chromatin instability and multipolarity to elicit cell death in TNBC,⁹ we hypothesized that the resistant lines evaded paclitaxel-induced lethality through stabilizing chromatin assembly and maintaining accurate mitosis, which however might be reversed by PRMT5 inhibition.

As anticipated, PRMT5 inhibition using either GSK591 or LLY283 treatment significantly downregulated the signaling pathways related to chromatin assembly and mitosis in three resistant clones of MDA-MB-436, including the nucleosome organization, chromatin assembly, and the mitotic sister chromatin segregation, while having no such effect on parental cells (Figures 2B, S3A, and S3B). These effects were recapitulated in Hs 578T cells and further validated by real time qPCR experiments in three resistant cell lines (Figures S3C, S3D, and S3E), underscoring the critical vulnerability of paclitaxel-resistant cells to PRMT5 inhibition.

To further determine whether the chromosome segregation defects observed in the transcriptome analysis led to mitotic phenotypes, cells treated with PRMT5 inhibitors were measured by immunofluorescence. While parental cells treated

with PRMT5 inhibitor LLY283 progressed normally through mitosis (Figure 2C), paclitaxel-resistant cells exhibited increased chromosome mis-segregation and multipolar spindle during mitosis upon LLY283 treatment (Figures 2D, 2E, and S3F). This resulted in cells with an incorrect number of chromosomes, potentially leading to the formation of aneuploidy. These observations demonstrate the essential role of PRMT5 in mitotic chromosome movements, specifically in paclitaxel-resistant cells. As previously reported, in parental TNBC cells, paclitaxel-mediated cell cytotoxicity was due to the induced chromosomal instability caused by chromosome mis-segregation and abnormal mitotic spindle (Figure S3G).²² However, the resistant cells maintained normal bipolar spindles throughout mitosis via the upregulated expression of chromosome assembly and segregation factors (Figures 2A and S3H). Altogether, these results suggest that targeting PRMT5 impaired the adaptive correction of chromosome mis-segregation in resistant cells, leading to chromosomal instability, the formation of multipolar spindles, and cell death.

PRMT5 regulates AURKB mRNA splicing in resistant cells

We further investigated the mechanism underlying the control of PRMT5 over mitotic chromatin stability. In addition to genes involved in mitosis, GSEA analysis identified downregulated RNA splicing pathways in paclitaxel-resistant cells following PRMT5 inhibitor treatment (Figures 3A and S4A–S4C; Table S4), corroborating previous reports on the established roles of PRMT5 in maintaining splicing fidelity.^{11,23,24} Hence, we hypothesized that altered mRNA splicing mediated by PRMT5 inhibitor treatment may result in cell cytotoxicity in paclitaxel-resistant cells. To test this hypothesis, alternative splicing events (ASEs) were analyzed in both parental and resistant cells in response to PRMT5 inhibition. As expected, treatment with PRMT5 inhibitor LLY283 induced significant increase in differential spliced events. Specifically, parental cells exhibited a total of 697 events, while the resistant cells displayed relatively higher probability of AS events (20RA: 1,166 events; 20RB: 907 events; 20RC: 711 events) (Figures 3B–3E). Similar effects were observed in GSK591 treatment (Figures S4D–S4G). Among these four types of ASEs, skipped exons and retained intron (RI) were predicted to have the most disruptive impact on their encoding proteins by VastDB analysis²⁵ (Figures S4H and S4I). In addition, resistant cells exhibited higher frequencies of RI events compared to parental cells in response to PRMT5 inhibition, suggesting that more affected proteins occur in resistant cells (Figure 3F).

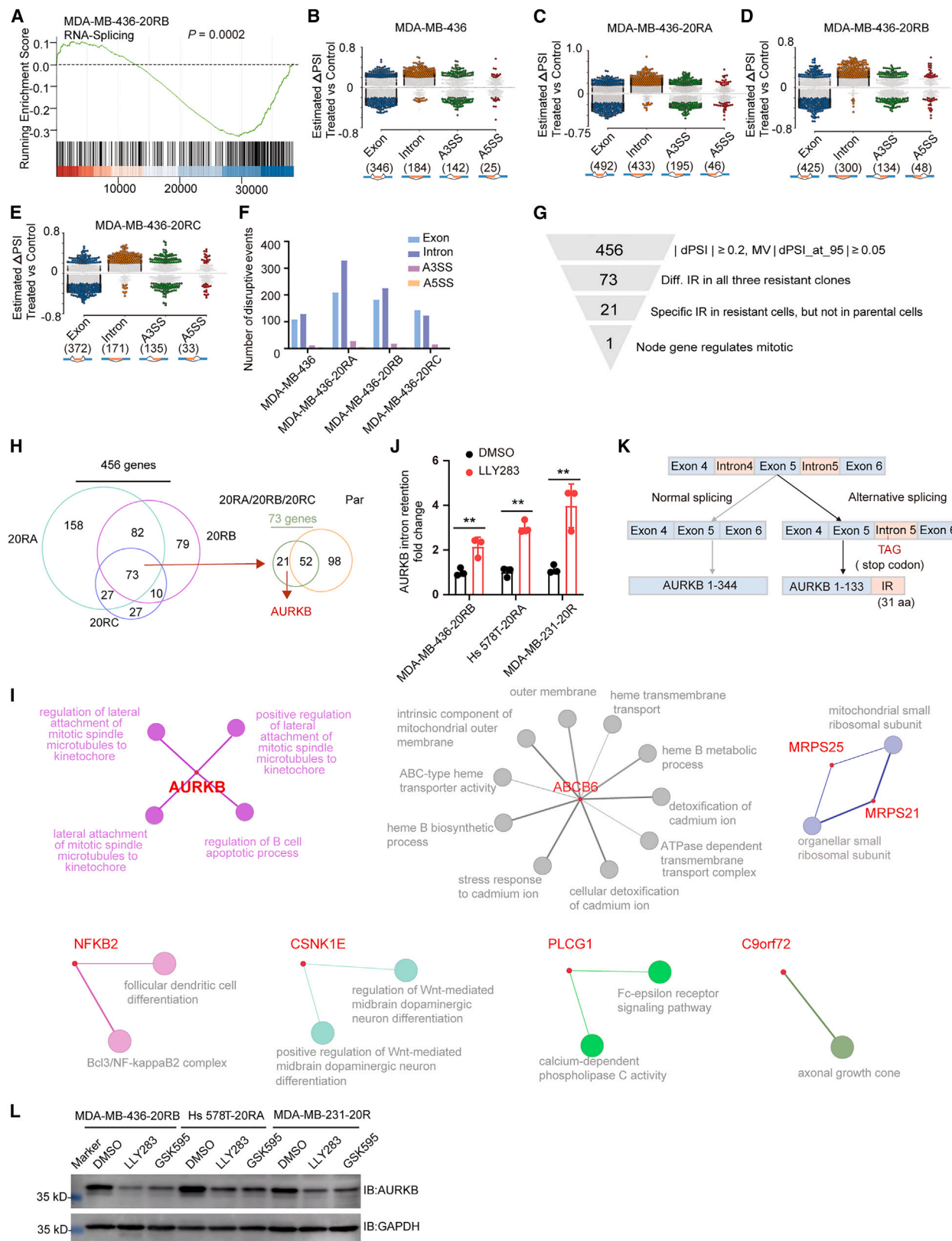
Figure 2. PRMT5 inhibition alters chromatin stability of paclitaxel-resistant cells

(A) Dot plot shows the top 20 pathways enriched in MDA-MB-436 paclitaxel-resistant cells compared to parental cells. NES, normalized enrichment score ($n = 2$, one-tailed Fisher's exact test).

(B) Lollipop plot shows the top 15 pathways enriched in MDA-MB-436 paclitaxel-resistant cells treated with LLY283 (1 μ M, 5 days) compared to DMSO control ($n = 2$, one-tailed Fisher's exact test).

(C and D) Immunofluorescence images of three parental (C) and paclitaxel-resistant (D) cell lines treated with 1 μ M LLY283 for 48 h, with α -tubulin antibody (CST, 2144S) staining for spindle and DAPI (Beyotime, C1102) staining for chromatin (scale bar, 10 μ m).

(E) Separate fluorescence quantification of normal or abnormal mitotic spindles in three parental and paclitaxel-resistant cells treated with 1 μ M LLY283 for 48 h (mean $\pm$ SD, Student's *t* test). Twenty cells covering four regions of interest were used for statistical analysis. The term "normal" refers to the process characterized by the proper formation and function of the spindle apparatus, which ensures that each daughter cell receives a complete and identical set of chromosomes. Abnormal mitosis refers to instances where defects in the spindle apparatus can lead to errors in chromosome segregation, resulting in cells with an incorrect number of chromosomes. **** $p < 0.0001$.



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To further elucidate the genes responsible for PRMT5-specific regulating mitosis in resistant cells, we focused on RNA splicing events occurring exclusively in resistant strains but not in parental cells, and these events should be consistent across different resistant clones. To achieve this, we initially analyzed 456 significant intron retention events induced by the PRMT5 inhibitor LLY283, identifying 73 genes with intron retention shared across different resistant clones. Subsequently, we filtered out 52 genes detected in parental cells from these 73 genes, resulting in 21 genes uniquely present in the resistant strains (Figures 3G and 3H; Table S5). Pathway enrichment analysis of these 21 genes revealed significant enrichment in chromosome stability and mitotic spindle attachment pathways, consistent with previous transcriptomic analysis results after PRMT5 inhibitor treatment. Notably, *AURKB* emerged as a central node within these pathways (Figure 3I). Given multiple studies reporting the essential role of *AURKB* in chromosome alignment and segregation during mitosis,²⁶ we directed our focus toward *AURKB* for further investigation.

In line with the RNA splicing analysis, real time qPCR confirmed that PRMT5 inhibition induced significant intron retention in *AURKB* across three paclitaxel-resistant cells (Figures 3J and S4J) but not in parental cells (Figure S4K). To investigate whether intron retention in *AURKB* affects protein expression, we first analyzed the *AURKB* sequence following LLY283 treatment. Our findings reveal that compared to the control group, PRMT5 inhibition induces retention of the 5th intron of *AURKB*, which harbors a premature termination codon, leading to premature termination of protein translation and thereby impeding the production of intact *AURKB* protein (Figure 3K). As expected, PRMT5 inhibition decreased *AURKB* protein levels in resistant cells (Figure 3L) but not in parental cells (Figures S4L and S4M).

To further validate this phenomenon, we separately overexpressed plasmids encoding *AURKB*^{WT} (344 amino acids) and *AURKB*^{IR} (164 amino acids) in paclitaxel-resistant cell lines and monitored *AURKB* protein expression using an *AURKB* antibody (ABclonal # A19539), which binds to the epitope spanning amino acids 1–100 at the N-terminal of *AURKB*. Our results indicate detectable levels of *AURKB* protein upon overexpression of *AURKB*^{WT}, whereas overexpression of *AURKB*^{IR} failed to yield an intact 39 kDa *AURKB* protein (Figure 4A). Although truncated *AURKB* proteins were generated by *AURKB*^{IR}, their functionality is compromised, as the kinase domain of *AURKB* resides at 77–327 amino acids at the C-terminus.²⁷ These findings indicate

that PRMT5 inhibition-induced intron retention in *AURKB* leads to premature translation termination, resulting in reduced *AURKB* protein expression.

To further validate the pivotal role of *AURKB* in PRMT5-mediated paclitaxel resistance, we conducted knockdown and rescue experiments targeting *AURKB* in three resistant strains. Our results demonstrated that *AURKB* knockdown significantly impaired the growth of resistant cells, while rescue with *AURKB*^{WT} restored cell growth; however, rescue with *AURKB*^{IR} failed to do so (Figures 4B and 4C). Additionally, we observed that overexpression of *AURKB*^{WT} in paclitaxel-resistant cells compromised their sensitivity to PRMT5 inhibitors, whereas overexpression of *AURKB*^{IR} did not significantly affect sensitivity to PRMT5 inhibitors (Figures 4D and S4N). These findings collectively suggest that PRMT5 modulates paclitaxel resistance by regulating intron retention in *AURKB*, leading to reduced *AURKB* protein levels and impaired growth of resistant cells.

AURKB inactivation has been reported to trigger cell-cycle arrest and BAK-mediated mitotic cell death.²⁸ Consistently, PRMT5 inhibitor treatment increased the BAK levels (Figures 4E, 4F, and S4O), thus resulting in cell apoptosis specifically in resistant cells (Figures 4G and S4P). Collectively, these results suggest that targeting PRMT5 alters RNA splicing and intron retention, leading to decreased protein levels of *AURKB* and cell death in paclitaxel resistant cells.

PRMT5 inhibition suppresses tumor growth in chemo-resistant PDX model

To assess the clinical relevance of PRMT5 in preventing chemoresistance in TNBC, we collected formalin-fixed paraffin-embedded (FFPE) tissue samples from patients who relapsed (pathological Miller-Payne scores: 1 or 2) after receiving eight cycles of neoadjuvant chemotherapy (NAC), which consists of a combination of anthracycline and cyclophosphamide followed by paclitaxel (AC-T) (Figure 5A). Subsequently, proteomic analysis was performed on these three pairs of FFPE samples obtained from patients who received this identical treatment protocol before and after NAC treatment. On average, we identified approximately 6,000 proteins in each sample (Table S6). Pathway enrichment analysis unveiled an enrichment of extracellular matrix-related pathways in the paclitaxel-resistant samples, which may be correlated with anthracycline and/or cyclophosphamide resistance.^{29,30} In agreement with previous transcriptome analysis, the enrichment of pathways associated

Figure 3. PRMT5 regulates *AURKB* mRNA splicing in paclitaxel-resistant cells

(A) GSEA plot of RNA splicing pathway in paclitaxel-resistant cells (MDA-MB-436-20RB) treated with 1 μ M LLY283 for 5 days ($n = 2$, one-tailed Fisher's exact test).

(B–E) Number of RNA alternative splicing events of MDA-MB-436 parental (B) or paclitaxel-resistant (C–E) cells treated with LLY283 (1 μ M, 5 days) compared to DMSO control. Numbers labeled below indicate estimated alternative splicing events.

(F) Number of predictive disrupted events in MDA-MB-436 parental and paclitaxel-resistant cells treated with LLY283 (1 μ M, 5 days) or DMSO control.

(G) Workflow for the identification of *AURKB*.

(H) Venn diagram illustrates 21 genes with retained introns specifically occurring in paclitaxel-resistant cells treated with the PRMT5 inhibitor (1 μ M LLY283, 5 days).

(I) Gene functional map of the 21 genes with retained introns. Red nodes indicate genes, other color nodes indicate cellular function, and clusters indicate functional classifications (Cytoscape 3.7.1).

(J) Expression changes of retained introns of *AURKB* gene in paclitaxel-resistant cells induced by LLY283 treatment (1 μ M, 5 days) compared to DMSO control (mean $\pm$ SD, $n = 3$, Student's t test).

(K) Sequence analysis of *AURKB* gene before and after LLY283 treatment.

(L) Immunoblots of *AURKB* in paclitaxel-resistant cells treated with PRMT5 inhibitors (LLY283, 1 μ M, 5 days; GSK595, 1 μ M, 5 days) or DMSO control. ** $p < 0.01$.

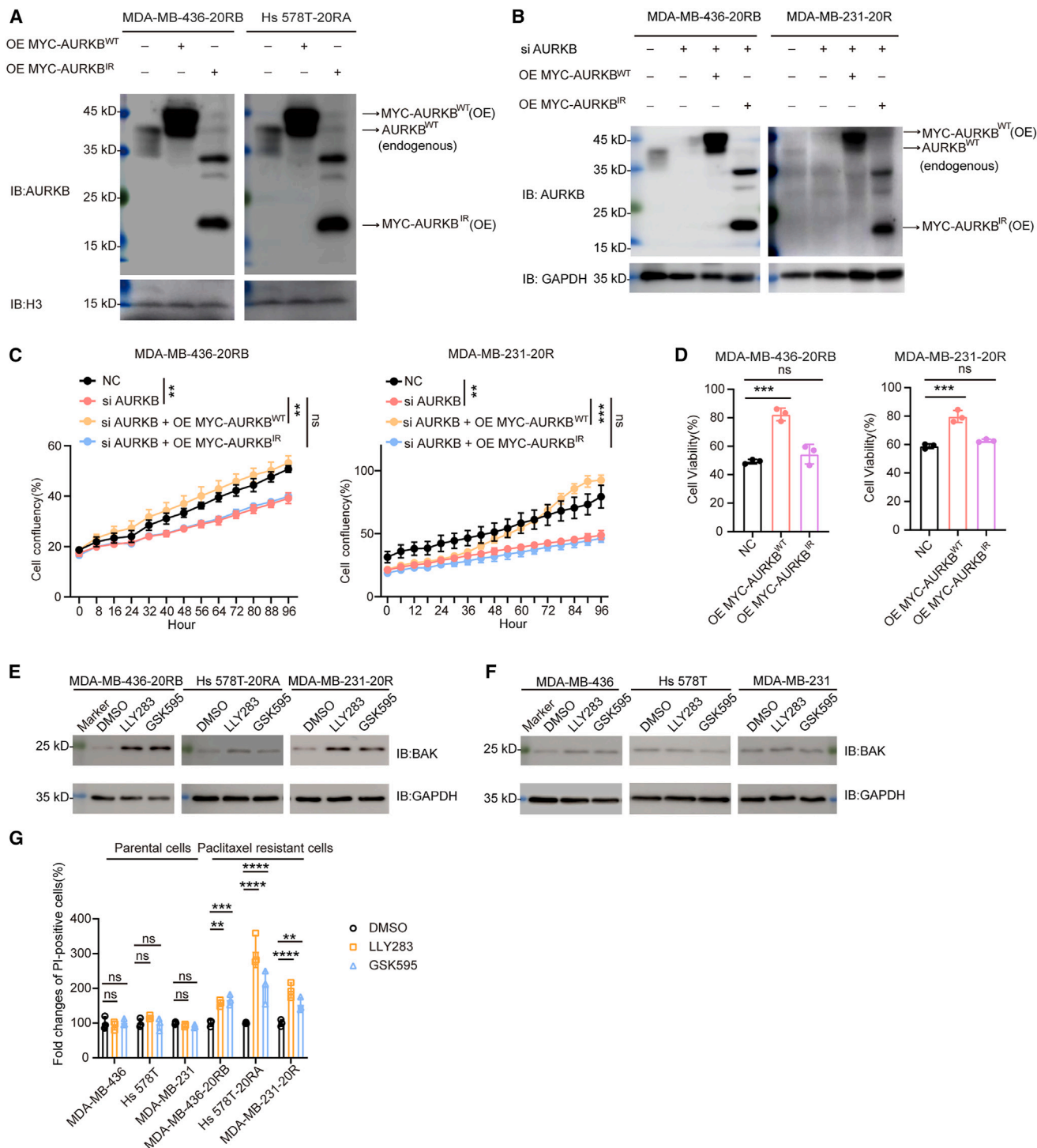


Figure 4. PRMT5 modulates paclitaxel resistance by regulating AURKB expression

(A) Immunoblots of AURKB in paclitaxel-resistant MDAMB-436-20RB, Hs 578T-20RA cells with overexpression of MYC-AURKB^{WT} or MYC-AURKB^{IR}.
 (B) Immunoblots of AURKB with indicated conditions.
 (C) Cell confluency of si-NC, si-AURKB, MYC-AURKB^{WT} and MYC-AURKB^{IR} rescued paclitaxel-resistant MDA-MB-436-20RB, MDA-MB-231-20R cells (mean $\pm$ SD, $n = 3$, Student's t test).
 (D) Cell viability of paclitaxel-resistant MDA-MB-436-20RB, MDA-MB-231-20R with overexpression of MYC-AURKB^{WT} or MYC-AURKB^{IR} treated with 1 μ M LLY283 for 5 days (mean $\pm$ SD, $n = 3$, one-way ANOVA).

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with chromatin stability was observed in the NAC-resistant samples (Figure 5B). In addition, genes associated with chromatin assembly and stabilization, including ANP32E, RANBP1, VPS4B, and NIPB, were upregulated in the NAC-resistant cells^{31–34} (Figure 5C).

To further validate the therapeutic efficacy of PRMT5 in paclitaxel-resistant TNBC, we investigated the impact of PRMT5 inhibitor LLY283 in a PDX model. This PDX model was derived from a patient who experienced relapse following treatment with the AC-T chemotherapy regimen. A daily oral gavage of 40 mg kg^{−1} LLY283 was administered to nude mice once the tumors became palpable (13–37 mm³). Monitoring of mice body weight measurement and histological analysis through tissue H&E staining revealed no detectable toxicity at this dosage (Figures S5A and S5B). Notably, PRMT5 inhibition significantly suppressed tumor growth as indicated by the reduced level of Ki-67 (Figures 5D, 5E, S5C, and S5D). *In vivo* confirmation of the on-target effect of PRMT5 was attained by the decreased in SDMA levels following LLY283 treatment (Figure 5F). Consistent with the cellular results, PRMT5 inhibition induced substantial intron retention in *AURKB* (Figure 5G) and reduced its protein level (Figure 5H), ultimately leading to cell apoptosis and tumor volume reduction, as evidenced by the elevated level of BAK protein (Figures 5I and 5J). Once again, GSEA analysis revealed that PRMT5 inhibition significantly impaired the pathways associated with chromatin stability, accurate chromosome segregation, and mitosis (Figure 5K). Collectively, these results provide compelling evidence for the anti-tumor effect of PRMT5 inhibition *in vivo*.

Type I PRMTs synergies with PRMT5 *in vitro* and *in vivo*

During the initial screening of epigenetic inhibitors, we observed that type I inhibitors effectively suppressed the growth of a paclitaxel resistant cell line (MDA-MB-436-20RB) at a high concentration of 5 μ M (Figure 1A), indicating the potential anti-proliferation effect of targeting type I PRMTs in TNBC. Mechanistically, our recent research along with others has demonstrated that type I PRMTs, similar to PRMT5, play crucial roles in RNA splicing.^{13,35} Additionally, studies have reported the synergistic effect of targeting PRMT5 and type I PRMT in suppressing cell growth in pancreatic adenocarcinoma, lung adenocarcinoma, and colon cancer tumor cells.^{36–38} Type I PRMTs are also known to share substrates with PRMT5.³⁹ Given these findings, we sought to explore the synergistic effect of PRMT5 inhibitors and type I inhibitors. As expected, the treatment of the type I PRMT inhibitor MS023 led to a decrease in *AURKB* protein levels by inducing intron retention in all three resistant cells (Figures 6A and 6B) but not in the parental cells (Figures S6A and S6B). The anti-proliferation effect was observed in cells with MS023 treatment, further validating the regulatory role of the splicing-*AURKB* axis in chemo-resistance (Figure 6C). Moreover, the combination of MS023 and PRMT5 inhibitors LLY283 or GSK595 led to a more significant reduction in *AURKB* protein levels and G2/M cell-cycle arrest, consistent with an increased level of intron retention (Figures 6D, 6E, S6C, and S6D). This indicated the potential synergistic effect of

MS023 and LLY283 in paclitaxel-resistant TNBC cells. To test this hypothesis, we treated the resistant cells with MS023 and LLY283 and used the Loewe additivity model to calculate the synergistic score considering the impact of both inhibitors on splicing, in which a score greater than 10 indicates synergy effect. The combination of MS023 and LLY283 showed strong synergy, including concentrations at which neither inhibitor had an effect on growth as monotherapy (Figure 6F).

We then evaluated the efficacy of this combination *in vivo*. Consistent with cellular results, both MS023 and LLY283 treatment showed an anti-tumor effect as evidenced by the decreased level of Ki-67 in the NAC-resistant PDX model as single agents (Figures 6G–6I, S6E, and S6F). Mice H&E tissue staining analysis and weight measurement indicated that no toxicity was detectable for the combination treatment (Figures S6G and S6H). In addition, compared to the monotherapy, the combination treatment significantly enhanced the inhibition of tumor growth, induction of intron retention (Figure 6J), and reduction of *AURKB* protein level (Figure 6K). The representative image and quantification analysis of the expression of the apoptosis marker (Figure S6I), and SDMA/ADMA markers (Figure S6J) illustrating the inhibition of PRMT5 and/or PRMT1 underscored the synergistic effect of PRMT5 inhibition and type I PRMTs inhibition. Together, these results collectively demonstrated that PRMT5 inhibition combined with type I PRMT inhibitor exerts a synergistic effect in suppressing tumor growth of resistant cells through the inhibition of *AURKB*-mediated mitotic signaling pathway.

DISCUSSION

Chemical probes are powerful tools for understanding human biology and treating disease.⁴⁰ They complement genetic knockdown/out methods and, due to their selectivity, are often used to identify proteins amenable to selective pharmacological targeting in specific disease models. In this study, with epigenetic probe screening in pairs of parental and resistant cells, we identified several epigenetic regulators, including bromodomain family proteins, histone methylation regulators of EZH2, G9a, and PRMTs, as potential targets in chemo-resistance TNBCs. The identification of these highly selective and potent agents suggests a reprogrammed epigenetic landscape in resistant cells, opening up avenues for the clinical resolution of paclitaxel drug resistance.⁴¹

Among the candidates identified in the screening, an inhibitor targeting PRMT5 has proven most effective in killing resistant cells, even at a low concentration of 1 μ M. This suggests the dependency of PRMT5 in paclitaxel-resistant cells, although the underlying mechanism remained to be elucidated. Consistent with a recent report showing that the anticancer effect of paclitaxel results from chromosome mis-segregation on multipolar spindles,¹¹ we found that TNBC-resistant cells evade chemotherapy by increasing the levels of *AURKB*-mediated mitotic chromatin stability. Subsequently treatment with PRMT5 and/or type I PRMT inhibitors decreases the protein level

(E and F) Immunoblots of BAK in paclitaxel-resistant (E) and their parental (F) cells treated with PRMT5 inhibitors (LLY283, 1 μ M, 5 days; GSK595, 1 μ M, 5 days) or DMSO control.

(G) Fold change of PI-positive cells in parental cells and their corresponding paclitaxel-resistant cells treated with PRMT5 inhibitors (LLY283, 1 μ M, 5 days; GSK595, 1 μ M, 5 days) compared to DMSO control (mean $\pm$ SD, $n = 3$, one-way ANOVA). ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, ns indicates not significant.

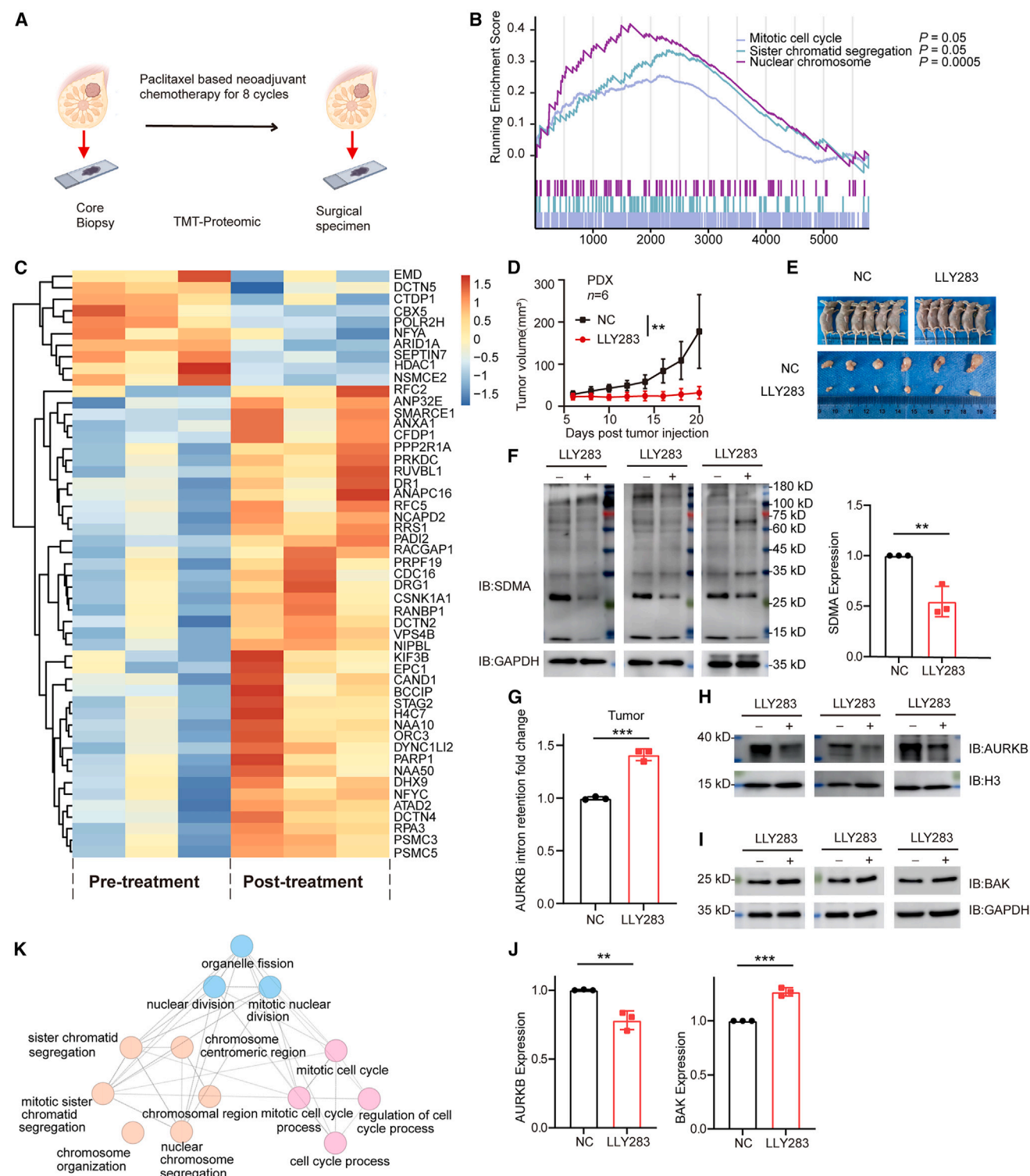


Figure 5. PRMT5i suppresses tumor growth in chemo-resistant PDX model

(A) Schematic diagram illustrates FFPE (Formalin-fixed paraffin-embedded) samples used for proteomics analysis.
(B) GSEA plot shows the enrichment pathways in chemo-resistant patient samples ($n = 3$, one-tailed Fisher's exact test).
(C) Heatmap shows expression levels of proteins associated with chromatin stability from patient samples before (pre-treatment) and after neoadjuvant chemotherapy (post-treatment).
(D) Tumor growth of NAC-resistant PDX models treated with LLY283 or control monitored till day 20 (mean $\pm$ SD, $n = 6$, two-way ANOVA).
(E) Images of mice and tumors at the endpoint of treatment.

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of mitotic regulator AURKB by inducing its intron retention, leading to chromosomal instability and, consequently, cell death, specifically in resistant cells.

It has been reported that cancer cells with elevated mitotic vulnerability, often accompanied by higher MYC proto-oncogene activity, rely on PRMTs to orchestrate splicing fidelity.^{42–44} Increased MYC activity has been shown to enhance dependency on PRMT5 in multiple cancer models. This may also be a contributing factor to the insensitivity of our TNBC parental cells to PRMT5 inhibitors.^{45–47} In addition, PRMT5 inhibition prevents the removal of detained intron of AURKB in MYC driven glioblastoma, underscoring our findings of selective sensitivity of PRMT5 inhibitors in resistant cells.⁴⁸ PRMT5 and PRMT1 may regulate AURKB's intron retention through methylation of RNA splicing factors, such as hnRNPA1,⁴⁹ SF3B2,⁵⁰ SRSF2,⁵¹ SmD1, and SmD3.^{52,53} Although their regulatory mechanisms are not yet fully understood, AURKB has been implicated in the development and progression of cancer and drug resistance through the activation of numerous mediators, including mitotic genes,^{16,54,55} suggesting it may serve as a common path to drug resistance.

We have sought to establish the clinical relevance of PRMTs-AURKB signaling in mediating acquired resistance to chemotherapy in TNBC. Analysis of clinical samples from patients who relapsed (pathological Miller-Payne scores: 1 or 2) after receiving eight cycles of NAC, validates that the acquisition of chemoresistance in TNBC patients is associated with increased levels of mitotic chromatin stability. Furthermore, PRMTs inhibition significantly suppressed tumor growth in a chemotherapy-resistant PDX model which belongs to basal-like 2 subtype. Multiple clinical trials are ongoing to evaluate the antitumor efficacy of inhibitors targeting PRMT5 or type I PRMTs in hematologic malignancies and advanced solid tumors, including breast cancer (NCT04676516, NCT06224387, NCT05528055, and NCT06130553). Although the clinical trial targeting type I PRMT with GSK3368715 has been terminated due to the limited benefit and a higher incidence of thromboembolic events during the dose escalation phase, the synergistic effect of low-dose PRMT5 and type I PRMT inhibitors may provide a potential therapeutic window with reduced side effects.⁵⁶

In summary, our chemical probes screening and mechanistic studies demonstrate that chemo-resistant TNBC acquired vulnerability to PRMT5 or type I PRMTs inhibitors, alone or in combination. Therefore, this work provides a rationale for initiating clinical trials to test PRMTs inhibitors in paclitaxel-refractory TNBC patients. Additionally, growing evidence has demonstrated that PRMTs also play an important role in the regulation of tumor immunity. Type I PRMTs can impact tumor antiviral innate immune response and tumor neoantigen expression in various cells such as TNBC and colorectal cancer by regulating

RNA splicing.^{13,35} PRMT5 can regulate tumor intrinsic immunity through cGAS/STING-IFN pathway in melanoma cells.⁵⁷ Given the roles of PRMTs in the immune system, the combination of immune-based therapies with PRMT5 or type I PRMT inhibitor demonstrates synergistic effects in preclinical samples and holds promise as a future treatment modality.^{35,57,58}

Limitations of the study

In this study, we found that inhibition of PRMT5 induces AURKB intron retention, thereby promoting cell death in paclitaxel-resistant TNBC cells. However, the mechanism by which PRMT5 inhibition causes AURKB intron retention remains unclear. Additionally, the potential of PRMT5 inhibitors to enhance immune therapy sensitivity suggests a need for further exploration of their function in paclitaxel-resistant TNBC. Moreover, we found that cancer cells displayed increased chromosomal stability during the development of resistance to paclitaxel. However, the precise biological pathways and molecular interactions that contribute to this enhanced stability are not yet fully understood. Furthermore, the clinical benefits of PRMT5 inhibitors for paclitaxel-resistant TNBC patients need to be further explored by clinical trials.

SIGNIFICANCE

Through chemical probe screening, we identified PRMTs as therapeutic targets in paclitaxel-resistant TNBC, highlighting the pivotal role of PRMTs in contributing to drug resistance within TNBC. Furthermore, we uncovered a mechanism by which PRMTs inhibitors effectively eliminate resistant cells. These inhibitors decrease the protein level of mitotic regulator AURKB by inducing intron retention, leading to chromosomal instability and, consequently, cell death, specifically in resistant cells. Moreover, our data demonstrate that type I PRMT and PRMT5 cooperatively regulate the growth of paclitaxel-resistant TNBC, thereby revealing the therapeutic potential of combining PRMT inhibitors in combating resistance. In summary, our findings provide valuable insights into overcoming drug resistance and suggest promising strategies for improved clinical outcomes in the treatment of paclitaxel-resistant TNBC.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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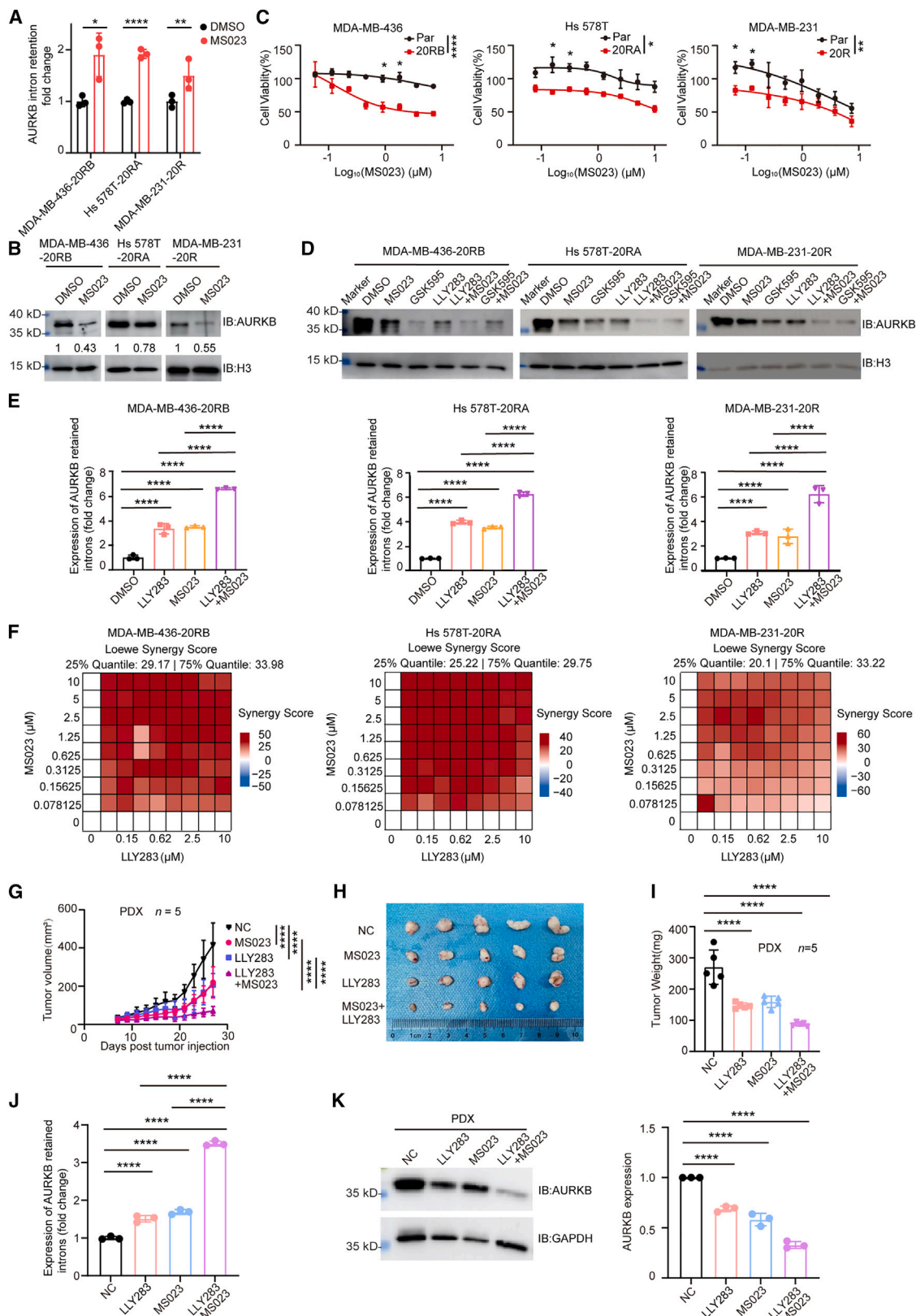
(F) Immunoblots of SDMA for PDX tumors with LLY283 or control treatment (left). Quantification of SDMA expressions normalized by GAPDH loading control (right, mean $\pm$ SD, $n = 3$, Student's t test).

(G) Real time qPCR of retained introns' level of AURKB mRNA in PDX tumors with LLY283 or control treatment, derived from half of the randomly selected mice in (D) (mean $\pm$ SD, $n = 3$, Student's t test).

(H and I) Immunoblots of AURKB (H) and BAK (I) for PDX tumors treated with LLY283 or control.

(J) Quantification of AURKB (H) and BAK (I) for PDX tumors treated with LLY283 or control. H3 or GAPDH was used as a loading control (mean $\pm$ SD, $n = 3$, Student's t test).

(K) Network of top 13 pathways enriched in NAC-resistant PDX treated with LLY283. Color nodes depict cellular functions ($n = 3$, one-tailed Fisher's exact test). ** $p < 0.01$, *** $p < 0.001$.



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- Data and code availability
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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.chembiol.2024.08.003>.

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AUTHOR CONTRIBUTIONS

K.J.Z. and J.W. coordinated and managed the overall project. Y.W., K.B.M., M.L., C.H.A., Q.W., and W.H.T., supervised the research. K.J.Z. planned the experiments, designed, and prepared figures and performed the bulk of the assays with J.W. S.Y.Z., and W.J.C. Y.S.J. L.Y.F., and L.G. performed the RNA-seq analysis. F.E.C. performed the splicing analysis. Real time qPCR and western blotting studies were performed by W.C.C., S.Y.Z. M.X.L., and X.Y.L. Z.J. and T.F. prepared TNBC FFPE samples. Mouse experiments were performed by K.J.Z. Y.Y., and J.W. Q.W., K.J.Z., J.W., and W.J.C. wrote the manuscript. All authors assisted in writing the manuscript, reviewed the final version, and approved the content and submission.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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Figure 6. Type I PRMTs inhibition synergies with PRMT5 inhibitor *in vitro* and *in vivo*

- (A) Levels of retained introns of *AURKB* in paclitaxel-resistant cells after MS023 treatment (1 μ M, 5 days) or DMSO control (mean $\pm$ SD, $n = 3$, one-way ANOVA).
- (B) Immunoblots of *AURKB* in paclitaxel-resistant cells after MS023 treatment (1 μ M, 5 days) or DMSO control. H3 was used as a loading control. Normalized band intensities are labeled below the images.
- (C) Cell viability curves of parental cells (MDA-MB-436, Hs 578T, and MDA-MB-231) and paclitaxel-resistant cells (MDA-MB-436-20RB, Hs 578T-20RA, and MDA-MB-231-20R) treated with MS023 for 5 days determined by the CCK-8 assay (mean $\pm$ SD, $n = 3$, Student's *t* test).
- (D) Immunoblots of *AURKB* in paclitaxel-resistant cells after treatments (LLY283, 1 μ M, 5 days; GSK595, 1 μ M, 5 days; LLY283 + MS023, 1 μ M + 1 μ M, 5 days; GSK595 + MS023, 1 μ M + 1 μ M, 5 days).
- (E) Real time qPCR results of retained introns of *AURKB* in paclitaxel-resistant cells treated with LLY283 (1 μ M, 5 days), MS023 (1 μ M, 5 days), the combination of LLY283 and MS023 (1 μ M, 1 μ M, 5 days), or DMSO control (mean $\pm$ SD, $n = 3$, one-way ANOVA).
- (F) Synergy map showing the dose-response of LLY283 and MS023 combination treatment in MDA-MB-436-20RB, Hs 578T-20RA, and MDA-MB-231-20R paclitaxel-resistant cells (SynergyFinder R package, Loewe additivity).
- (G) Tumor growth curves of a PDX model treated with LLY283, MS023, the combination of LLY283 and MS023, or control ($n = 5$, two-way ANOVA).
- (H) Images of tumors at the endpoint after 27 days of treatment.
- (I) Tumor weight of a PDX model treated with LLY283, MS023, the combination of LLY283 and MS023 or control at end of the experiment (mean $\pm$ SD, $n = 5$, one-way ANOVA).
- (J) Real time qPCR of retained introns' levels of *AURKB* in PDX tumors treated with LLY283, MS023, the combination of LLY283 and MS023 or control (mean $\pm$ SD, $n = 3$, one-way ANOVA).
- (K) Representative immunoblots of *AURKB* for NAC-resistant PDX in treated groups and the control group originate from a random selection of mice out of the five utilized in (G) (left). *AURKB* expressions were quantified and normalized with GAPDH loading control derived from a random selection of three mice out of the five utilized in (G) (right, mean $\pm$ SD, $n = 3$, one-way ANOVA). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti- PRMT5	Cell Signaling Technology	Cat# 79998S; RRID: AB_2799945
Anti-SDMA	Cell Signaling Technology	Cat#13222S; RRID: AB_2714013
Anti-ADMA	Cell Signaling Technology	Cat# 13522S; RRID: AB_2665370
Anti-GAPDH	Cell Signaling Technology	Cat# 2118; RRID: AB_561053
Anti-H3	Cell Signaling Technology	Cat# 4499S; RRID: AB_10544537
Anti-AURKB	ABclonal	Cat# A19539; RRID: AB_2862656
Anti-BAK	ABclonal	Cat# A0204; RRID: AB_2757018
Anti-ALDH1A1	HUABIO	Cat# R1706-2; RRID: AB_3073429
Anti-CD44	HUABIO	Cat# ET1609-74; RRID: AB_3069883
Anti-Ki-67	Abcam	Cat# 4203-1; RRID: AB_765010
Secondary antibodies	Cell Signaling Technology	Cat# 7076P2 or 7074P2; RRID: AB_330924 or RRID: AB_2099233
Anti- α -Tubulin	Cell Signaling Technology	Cat# 2144S; RRID: AB_2210548
Bacterial and virus strains		
Rosetta2	Sigma	Cat# 70954
Biological samples		
Human TNBC tumor tissue FFPE	Xiangya Hospital	N/A
Patient-derived xenografts (PDX)	Zhejiang cancer hospital	N/A
Mouse breast cancer tumor tissues	This paper	N/A
Chemicals, peptides, and recombinant proteins		
GSK2801	Sigma	Cat# SML0768
BAZ2-ICR	Sigma	Cat# SML1276
PFI-1	Sigma	Cat# SML0352
JQ1	Sigma	Cat# SML1524
JQ1N	Sigma	Cat# SML1525
BAY-299	Cayman	Cat# 19777
I-BRD9	Sigma	Cat# SML1534
LP99	Cayman	Cat# 17661
BI-9564	Sigma	Cat# SML1655
PFI-4	Sigma	Cat# PZ0307
OF-1	Sigma	Cat# SML1184
NI-57	Sigma	Cat# SML1486
SGC-CBP30	Sigma	Cat# SML1133
I-CBP112	Sigma	Cat# SML1134
PFI-3	Sigma	Cat# SML0939
TP-472	Sigma	Cat# SML1862
SGC0946	Sigma	Cat# SML1107
A-395	Cayman	Cat# 20257
A-395N	Sigma	Cat# SML1879
UNC1999	Sigma	Cat# SML0778
GSK343	Sigma	Cat# SML0766
GSK126	Sigma	Cat# 5.0058
UNC0642	Sigma	Cat# SML1037
A-366	Sigma	Cat# SML1410
PFI-2	Sigma	Cat# SML1408

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
S-PFI-2	Sigma	Cat# PZ0312
BAY-598	Sigma	Cat# SML1603
A-196	Sigma	Cat# SML1565
OICR-9429	Sigma	Cat# SML1209
MS023	Cayman	Cat# 34786
MS094	Sigma	Cat# SML2548
TP-064	Cayman	Cat# 20256
SGC707	Sigma	Cat# SML1242
GSK591	Cayman	Cat# 18354
UNC1215	Sigma	Cat# SML0662
UNC1079	Cayman	Cat# 20566
GSK-J4	Sigma	Cat# SML0701
GSK-LSD1	Sigma	Cat# SML1072
IOX2	Sigma	Cat# SML0652
GSK484	Sigma	Cat# SML1658
GSK864	Sigma	Cat# SML1757
LLY283	MCE	Cat# HY-107777
GSK595	MCE	Cat# HY-18665
DAPI	Thermo Fisher	Cat# D21490

Critical commercial assays

FastPure cell/tissue total RNA isolation Kit V2	Vazyme	Cat# RC112
HiScript II one-step qRT-PCR CYBR Green Kit	Vazyme	Cat# Q221
Cell Counting Kit-8	Biosharp	Cat# BS350B
Bradford Protein Assay Kit	Biosharp	Cat# BL524A
Cell Cycle and Apoptosis Analysis Kit	Beyotime	Cat# C1052

Deposited data

GO gene sets	MsigDB	MsigDB: https://www.gsea-msigdb.org/gsea/msigdb/
Original data of statistical analyses and uncropped gels	This paper; Mendeley data	Mendeley Data: https://data.mendeley.com/datasets/x29rjbpk8r/1
Raw and analyzed data of transcriptome sequencing	This paper	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE264630 ; GEO: GSE264630
Proteomics data	This paper; ProteomeXchange	http://proteomecentral.proteomexchange.org/cgi/GetDataset?ID=PX053336 ; PRIDE: PX053336

Experimental models: cell lines

MDA-MB-436	ATCC	Cat# Bio-73324; RRID:CVCL_0623
MDA-MB-436-20RA	This paper	N/A
MDA-MB-436-20RB	This paper	N/A
MDA-MB-436-20RC	This paper	N/A
Hs 578T	ATCC	Cat# Bio-54123; RRID:CVCL_0332
Hs 578T-20RA	This paper	N/A
Hs 578T-20RB	This paper	N/A
MDA-MB-231	ATCC	Cat# Bio-73091; RRID:CVCL_0062
MDA-MB-231-20R	This paper	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: organisms/strains		
Mouse: Balb/c nude mice	Laboratory Animal Center, Hangzhou medical college	RRID:RGD_2312499
Oligonucleotides		
See Table S7 for the detailed sequences	N/A	N/A
Recombinant DNA		
PCMV-PRMT5 (human) –3 x Myc Neo	MiaoLing	Cat# P51687
pLV3-CMV-AURKB (human) –3 x Myc Puro	MiaoLing	Cat# P48447
PLV3-AURKB Inclusion - (human) –3 x Myc	This paper	N/A
Software and algorithms		
R 4.3.0	N/A	https://cran.r-project.org/src/base/R-4/ ; RRID: SCR_001905
ImageQuant TL v8.2.0	Cytiva	https://www.cytivalifesciences.com/en/us/shop/protein-analysis/molecular-imaging-for-proteins/imaging-software/imagequant-tl-10-2-analysis-software-p-28619 ; RRID: SCR_014246
ImageJ (1.53K)	NIH	https://imagej.net/ij/ ; RRID: SCR_003070
GraphPad Prism 9	GraphPad Software	https://www.graphpad.com/ ; RRID: SCR_002798
STELLARIS 8 Confocal Microscope	Leica	https://www.instrument.com.cn/netshow/SH100826/C490873.htm ; RRID: SCR_024664
Incucyte SX5 Live-Cell Analysis Systems	Satorius	https://www.instrument.com.cn/netshow/SH100266/C401167.htm ; RRID: SCR_019874
Fully automatic tissue dyeing and sealing all-in-one machine	SAKURA	https://www.bio-equip.com/show1equip.asp?equipid=3317978
Real-time PCR	Roche	https://china.guidechem.com/trade/pdetail25641969.html ; RRID: SCR_018626
Fully automatic pathological slice scanning	Olympus	https://ex.chinadaily.com.cn/exchange/partners/82/rss/channel/cn/columns/vyuat/stories/WS5dfb3c22a31099ab995f28c1.html
Code used in this study	This paper	Github: https://github.com/chonga2022/PRMT5-paclitaxel-resistant-TNBC ; Zenodo: https://doi.org/10.5281/zenodo.12754481

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Qin Wu (wuqin@ibmc.ac.cn).

Materials availability

Plasmids and cell lines generated in this study will be made available on request, but we may require a payment and/or a completed Materials Transfer Agreement if there is potential for commercial application.

Data and code availability

- The transcriptome sequencing information generated in this study has been uploaded in the NCBI Gene Expression Omnibus (GEO) and can be queried through the GEO: GSE264630. The proteomics data have been deposited to the ProteomeXchange

Consortium via the iProX partner repository with the dataset identifier PRIDE: PXD053336. All repetitive trials and raw data are deposited at Mendeley Data: <https://data.mendeley.com/datasets/x29rjbpk8r/1>. All data are publicly accessible as of the date of publication.

- All code used is openly available at Zenodo: <https://doi.org/10.5281/zenodo.12754481>, can be found on GitHub: <https://github.com/chonga2022/PRMT5-paclitaxel-resistant-TNBC>.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell lines

TNBC cell lines were obtained from ATCC and cultured according to culture conditions. Authentication information is available at the vendor website. MDA-MB-436, MDA-MB-231, and Hs 578T cell lines were cultured in DMEM medium (Gibco, C11995500BT) containing 10% FBS (BI, 04-001-01ACS) and 1% penicillin/streptomycin (Biosharp, BL505A). Cells were cultivated at 37°C in an incubator containing 5% CO₂ (Thermo Fisher, 50145515), and cells were passaged no more than 25 times. The sex of cell lines is female.

Animals

The animal experiments were approved by Hangzhou Institute of Medicine Animal Experiment Ethics Committee (Approval ID: 2023R-0053). The female BALB/c nude mice used in this study were all around 6 weeks old. Mice were housed in pathogen-free and controlled conditions in individually ventilated cages and fed an irradiated soy-free diet and autoclaved tap water *ad libitum*. Paclitaxel-resistant PDX was cut into the size of a small grain and implanted on the upper right side of female BALB/c nude mice. When tumors grew to measurable sizes, they were randomly grouped according to experimental needs. During the first experiment, female BALB/c nude mice were given LLY283 at 40 mg kg⁻¹ every other day. The experiment was stopped 20 days following the treatment. During the second experiment, female BALB/c nude mice were given LLY283 at 20 mg kg⁻¹ daily and MS023 at 30 mg kg⁻¹ every other day.

Human subjects

The use of human tissue FFPE was approved by the Medical Ethics Committee of Xiangya Hospital Central South University, with informed consent obtained from all participants (Approval ID: 202211708). The human tissue FFPE samples used for proteomic analysis were obtained from female patients who relapsed (pathological Miller-Payne scores: 1 or 2) after receiving eight cycles of neoadjuvant chemotherapy (NAC), which consists of a combination of anthracycline and cyclophosphamide followed by paclitaxel (AC-T).

METHOD DETAILS

Plasmids

PCMV-PRMT5 (human) –3 × Myc Neo and PLV3-CMV-AURKB (human) –3 × Myc Puro were purchased from MiaoLing under item numbers P51687 and P48447. For PLV3-AURKB Inclusion - (human) –3 × Myc construction, PCR was performed using primers that contained XhoI and EcoRI restriction enzyme sites to amplify the *AURKB* intron-retention region. The PLV3 vector was digested with XhoI and EcoRI at 37°C for 4 h. Following digestion, the vector was purified by gel extraction to remove excess enzymes and non-ligated DNA. The PCR-amplified fragment and the purified, digested vector were ligated at 16°C overnight using T4 DNA ligase. The ligation product was transformed into competent *Escherichia coli* cells. Transformed colonies were selected on LB agar plates containing puromycin, an antibiotic used to screen for successful plasmid integration. Sequences were confirmed by DNA sequencing.

Epigenetic chemical screening

Compounds were purchased from Cayman, Millipore Sigma, MedChemExpress, or donated by the SGC, Canada. Chemical purity was verified to be over 99%. For screening, each well was plated with 1000 TNBC cell on 384-well plates, and after cells adhered to the plates, compounds dissolved in DMSO were added to reach a final concentration of 1 μM or 5 μM. Each compound and concentration were done in our replicates, with DMSO serving as a control. Cells were then placed in an incubator and treated with the added compounds for 5 days and cell proliferation was analyzed using a live cell analysis system (Sartorius, Incucyte S3) with a 10× objective, according to the manufacturer's instructions.

Cell viability assay

TNBC cells were seeded into 384-well plates at a density of 1,000 cells per well or into 96-well plates at a density of 4,000 cells per well. After cells adhered to the plates, small molecule inhibitors were added and treated for 5 days. Cell Viability Assay Reagent CCK8 (Biosharp, BS350B) was added to each well to reach a 10% final concentration. After gently shaking the plates and incubating for 1 h at 37°C, protected from light, the values were measured in a multimode microplate reader (TECAN, Spark).

Western blot

Cells or tissues were collected and lysed on ice with lysis buffer (Beyotime, p0013) adding 1× protease inhibitor cocktail (Roche, 5056489001), 1× phosphatase inhibitor cocktail (Beyotime, P1045), and Benzamide (Beyotime, D7121-25KU) for 10 min. Proteins were quantified with BSA protein quantification kits (Beyotime, p0010). 10–20 µg of protein lysate extract were ran on 10%–12% gradient gels and transferred to methanol-activated PVDF membranes at 80V for 2 h. Membranes were blocked with 5% skim milk for 1 h at RT. Primary antibodies used for membrane staining were anti-PRMT5 (CST, 79998S; 1:1000), anti-SDMA (CST, 13222S; 1:1000), anti-GAPDH (CST, 5174S; 1:1000), anti-H3 (CST, 4499S; 1:1000), anti-ALDH1A1 (HUABIO, R1706-2; 1:1000), anti-CD44 (HUABIO, ET1609-74; 1:1000), anti-AURKB (ABclonal, A19539; 1:1000), anti-BAK (ABclonal, A0204; 1:1000). Membranes were stained with secondary antibodies (CST, 7076P2 or 7074P2) for 1 h at RT and imaged (AMERSHAM, IMAGE QUANT 800).

Real time qPCR

RNA was extracted from cells or tissues using an RNA extraction kit (Vazyme, RC101). After cDNAs were transcribed (Yeasen, 11119ES60), gene expression was quantified by SYBR Green Quantification Kit (Bio-rad, 1725124) on the CFX Maestro (Bio-Rad). The amount of cDNA template was determined according to the instructions for the SYBR Green kit. The mixture was added to a white PCR reaction tube and placed in a CFX Maestro. The PCR reaction ran with the following conditions: 5 min at 95°C, 35 cycles of 45 s at 95°C, 30 s at 57°C, 40 s at 72°C, 7 min at 72°C, and 4°C indefinitely. Data were collected on the Bio-Rad CFX96 and analyzed by the Bio-Rad CFX Manager. All experiments were performed in at least three biological replicates. The primers used for this study can be found in [key resources table](#).

Paclitaxel-resistant cell lines generation

MDA-MB-436, Hs 578T, and their resistant lines were previously established.¹⁰ As for constructing the MDA-MB-231 resistant line, we obtained parental MDA-MB-231 cells from Fudan University. MDA-MB-231 were cultured in DMEM medium containing 10% FBS (BI,04-001-01ACS) and 1% penicillin/streptomycin (Biosharp, BL505A). The cells were first treated with 10 nM Paclitaxel, and surviving cells continued to be cultured. We then increased the concentration to 20 nM Paclitaxel, and the surviving cell was maintained with 20 nM Paclitaxel until growth is stable. Then we performed cell proliferation assays in all parental and resistant lines with increasing doses of Paclitaxel to measure IC₅₀ values. Cells were considered drug-resistant cells when the resistance index reached 5.

Immunofluorescence

TNBC cell lines were seeded in confocal culture dishes and treated with 10 nM paclitaxel for 36 h. After fixing the cells with 4% paraformaldehyde fixative (prepared using PBS at pH 7.4), cells were permeabilized by adding 0.2% Triton X-100. After blocking cells with 3% BSA for 30 min, primary antibody (α -Tubulin, CST, 2144S) was incubated overnight at 4°C. The next day, secondary antibodies (Goat Polyclonal to Goat anti-rabbit IgG, Huabio, HA1121) were incubated for 1 h at RT, and DAPI (Beyotime, C1102) was used for staining chromosomes, and finally, chromosomes and spindle were observed by confocal microscope (Leica, stellaris 8).

HE staining

Mice heart, liver, spleen, lungs, and kidney were removed and fixed with 4% paraformaldehyde fixative for 5 days at room temperature. Then placed in an embedding box for tissue dehydration and paraffin embedding. The paraffin sections were trimmed until the tissue surface reached a representative section (Leica, BOND RXM). Then tissues were cut to 5 µm thickness, dewaxed in warm water, and fixed onto microscope slides. After eluted with different concentrations of alcohol, hematoxylin and eosin staining were performed. After dehydration with different concentrations of alcohol, the slides were sealed with resin. Finally, pictures were taken using the tissue observation apparatus (Olympus, VS200).

Preparation and dispensing of compounds

MS023 was administered by intraperitoneal injection. 62.1 mg of MS023 was weighed, and placed into a beaker with 0.368 mL NMP (Macklin, C12412679), 1.474 mL captisol (20% w/v, Aladdin, J2126302), 1.474 mL of PEG-400 (MCE, 109420), and 4.054 mL of saline. Then the mixture was vortexed and shaken continuously for 2 min 200 mg of LLY283 (MCE, HY-107777) was dissolved in 4 mL of DMSO and dispensed as required. LLY283 vehicle solution contains 0.4 g 1% HEC, 0.1 mL 0.25% Tween80, 0.025 g 0.05% Anti-foam, and 35.9 mL PBS. LLY283 was then diluted with the vehicle solution and administered to mice orally.

Bioinformatic analysis

Gene set enrichment analysis. We obtained the transcriptome data of parental and paclitaxel-resistant cell lines (n = 2), chemo-resistant TNBC patients (n = 3) and PDX model (n = 3) through RNA-seq and processed the data with the Kallisto pipeline, genes were ranked based on stat value from R package DESeq2 (v1.34) or edgeR (v3.18). GO gene sets were downloaded from MsigDB (<https://www.gsea-msigdb.org/gsea/msigdb/>). The R package clusterProfiler (v3.18) was used to generate the GSEA results, and the R package enrichplot (v3.18) was used to create pathway enrichment plots. Sequencing raw data have been uploaded to the Gene Expression Omnibus (GEO) public repository (accession number GEO: GSE264630). All code used for gene set enrichment analysis is openly available at Zenodo : <https://doi.org/10.5281/zenodo.12754481>.

Sequence alignment, transcript reconstruction, and alternative splicing events (ASE) identification

FastQC was used to evaluate sequencing quality of each library⁵⁹ followed by alignment to human genome GRCh38.p5, using STAR aligner v2.6.1 (0.11.8)⁶⁰ and v24 from GENCODE's comprehensive gene annotation as a ref.⁶¹ To ensure that most of the aligned reads were uniquely mapped and properly paired, the BAM files were evaluated with samtools flagstat.⁶² Aligned files from all samples were used as input for the Stringtie v.1.3.4 pipeline for the assembly of transcripts found in the RNA-seq data.⁶³ The four classes of ASEs were identified by MATS v3.2.5 in a single-sample mode for exon skipping, intron retention, MXEs, and alternative 3' and 5' splice sites.⁶⁴ Intron retention events were identified using IRFinder v.1.2.4 for single samples.⁶⁵ For all classes, statistically significant events were selected based on a combination of parameters: *p* value ≤ 0.05 ; at least 1 read supporting the inclusion event; at least 1 read supporting the exclusion event; at least 10 total reads supporting the event; and ΔPSI (Treated – Control) ≥ 0.10 .

ASE characterization

Networks for protein-coding genes containing retained introns were analyzed by Cytoscape networks version 3.7.1.⁶⁶ Additionally, alternatively spliced events were cross-examined with VastDB²⁵ (hg38 assembly) using bedtools intersect to identify common elements based on their genomic coordinates, requiring a 95% reciprocal overlap and same strandness to be considered the same event.

FFPE proteomic analysis

The use of human tissue FFPE was approved by the Medical Ethics Committee of Xiangya Hospital Central South University, and patient consent was obtained (Approval ID: 202211708). Five to 10 FFPE slides (5 μm) were used for simultaneous protein isolation. The FFPE samples were dewaxed and rehydration three times with xylene and one time with 100%, 95%, 75%, and 50% ethanol. Every sample was mixed thoroughly with 200 μL lysis buffer (4% SDS, 1% Protease inhibitor cocktail [Sigma]) and put on ice for 15 min. Then the samples were ultrasonicated for 10 min (3 s-on, 3 s-off). The lysis samples were heated at 95°C, 750 rpm for 1 h. The concentration was measured by BCA method. The samples were added with 2 μL 500 mM DTT and kept at 56°C for 1 h. Then 20 μL 500 mM IAM was added and the samples were kept at room temperature, blocked from light for 45 min. Up to 60 μg protein was purified with SP3 beads. 20 μL digestion buffer (50 mM NH_4CO_3 , 50 mM CaCl_2 , 2.4 μg trypsin) was used to resuspend the beads. The proteins were digested overnight at 37°C and 1 μg trypsin was further added to digest for another 3 h at 37°C. The peptides were then purified by SP3 beads and eluted with 20 μL 2% iRT standards (Biognosys, Schlieren, Switzerland). The data from lysed peptide samples were acquired through Orbitrap Q Exactive HF mass spectrometry (Thermo Fisher Scientific, Waltham, MA) accompanied by a Thermo Scientific UltiMate 3000 UHPLC system. The digested peptides were ionized under 2 kV and introduced into mass spectrometry under a data independent acquisition (DIA) mode. Ions with *m/z* ranging from 300 to 1,500 were acquired by Orbitrap mass analyzer at a high resolution of 60,000. Precursor ions were fragmented with higher energy collision dissociation (HCD) with a normalized collision energy of 32%. The data were searched against the human UniProt database (20,365 sequences) using Spectronaut software (version 14.5.200813.47784). The library generation with data applied the default settings with trypsin/P digest rule, high protein, peptide confidence level, and FDR of 0.01. Fold change of 1.5 times, and a *t* test *p* value of 0.05 were set as the cut-off value for differential proteins. Proteins with an area under curve value greater than 0.8 were analyzed for associations with clinical outcomes. Unsupervised clustering and heatmap were constructed using custom R scripts and R packages.

siRNA transfection

Cells were plated in 6-well plates at 3.0×10^5 cells/well, and cultured at 37°C, 5% CO_2 incubator for 24 h before transfection. SiRNA and lipo2000 were diluted into Opti-MEM separately (Gibco, 31985062). Each mixture was pipetted gently and incubated at room temperature for 5 min. We then mixed the siRNA and lipo2000 mixture together and incubated for 30 min at room temperature. Cell media were replaced with 1.5 mL serum-free media, and the mixture was added dropwise. After cultured in the incubator (37°C, 5% CO_2) for 7 h, media were replaced with 2 mL complete media containing 10% FBS and 1% antibiotic. Cell confluency was observed over a 72 h period using a live cell analysis system (Sartorius, Incucyte S3) with a 10 $\times$ objective. Cells were collected at 72 h time points for WB. siRNAs were provided by General Bio Inc.

QUANTIFICATION AND STATISTICAL ANALYSIS

We used ImageJ to quantify the protein expression level of the bands in the gel image. We normalized the quantified protein levels against those of the control group to generate expression ratios. The raw, unprocessed data that underpin these quantifications are provided in the [supplemental information](#). For the gene expression assays, mean light intensity data were normalized as a ratio of the sample, with the raw data shown in the [supplemental information](#). All experiments were reproducible. The log rank test was used to analyze differences between groups. GraphPad Prism9 was used for calculating IC_{50} values, and statistical analysis. Significance was calculated using Student's *t* test or multiple comparisons. A *p* value < 0.05 was considered significant. * indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.001$, and **** indicates $p < 0.0001$. Synergy finder was used to generate Loewe synergy scores. For cellular proliferation assay, cell viability assay, real time qPCR assay, apoptosis assay, and Western blot assay, "n" indicates the number of biological replicates. The images presented are representative of three independent biological replicates. For animal experiments, "n" refers to the number of animals per group. For FFPE proteomic analysis, "n" indicates the number of patients. Data are presented as mean $\pm$ standard deviation. The statistical test methods were shown in the figure legends.